CONCLUSION

To us, the advent of AI is the start of an odyssey of the spirit as much as it is an expedition of logic and truth. We cannot properly consider, let alone prepare for, coexistence and coevolution with not-human life by relying on rationality alone—whether that of a machine or that of a human. Rather, something more is required—something fundamentally human.

It has long been hypothesized that our universe is something akin to an ancient game of chess—continuously played by entities older and larger than our present observable reality. The longer we have observed the movements on the board, the likelier it has become that we would eventually catch on to a few rules of play. After watching long enough, we might even begin to play ourselves. The move from passive observation to active participation is not a leap of logic. Translating principles into action is always a leap of faith.

Asked about his religious beliefs, Albert Einstein responded:

> We are in the position of a little child, entering a huge library, whose walls are covered to the ceiling with books in many different tongues. The child knows that someone must have written those books. It does not know who or how. It does not understand the languages in which they are written.
>
> The child notes a definite plan in the arrangement of the books, a mysterious order, which it does not comprehend, but only dimly suspects.
>
> That, it seems to me, is the attitude of the human mind, even the greatest and most cultured, toward God.
>
> We see a universe marvelously arranged, obeying certain

> laws, but we understand the laws only dimly. Our limited minds cannot grasp the mysterious force that sways the constellations.[1]

Our minds remain childlike with respect to God, our world, and now our newest creations. Understanding what we have wrought, in a logical and spiritual sense, will be an essential step toward maturity as a species. But we must also make a leap, the leap of faith, to move beyond examination and into intervention. This will require us to act in a state of uncertainty: the quagmire of human leaders since the dawn of time. Action has never meant the privilege of total control—indeed, quite the opposite. Nor will it do so in the age of AI; we do not expect to know our future destiny more than we have known it in prior epochs.

Our lack of control does not require us to abandon reason or, worse still, to give up our investment, and our inclination to act, in the real world. However, as humans enter a new cycle that involves direct partnership with AI, the particular dynamics spelled out in this volume will encourage us to lean into new and old forms of inquiry alike. Our success will depend on a commitment to defining and acting upon our moral convictions. That will require steadfast courage—and a consistent strategy—as new truths alter preexisting conceptions.

In fact, it is moral purposes that encourage our continued progress. Moreover, it is the very existence of an essential substratum of human morality that enables humanity's transcendence of the dichotomy between human control (tactical, at best) and human benefit (abundant, by any definition). It is a fundamental belief in the reality of human dignity that allows us, the authors of this book, at once to recognize the promise of human-AI alignment and to accept the necessity of faith in navigating the years ahead as science advances and reveals ever greater mysteries.

Despite our many shared human qualities, we cannot expect unity in the choices that lie ahead. What some see as an anchor to steady ourselves in the storm, others see as a leash holding us back.

What some praise as necessary steps toward a pinnacle of human potential, others see as a headlong rush into an abyss.

In this case, instinctive emotional divergences—and the subjective lines that are drawn by all parties—will create an unpredictable and combustible situation. Increasingly stark positions of potential "winners" and "losers" will intensify the pressure of these circumstances. The fearful will slow their own development and sabotage that of others. The overconfident will disguise their powers and, in secret, speed up their work. The timeline of coming crises will be accelerated beyond prior human experience; quickly, we will be engulfed, and it is not clear whether or how we will survive.

Might AI cause these coming crises, and then act as our savior—manufacturing problems that only it can solve, if only to prove its necessity and to remind us of our dependency? Again we return to the dilemma that has motivated much of this volume: the excruciating choice between control and utility, between the comfort of the historically independent human and the possibilities of an entirely new partnership.

That choice is difficult and necessary. It is also solvable provided one assumes in humanity a true, definable, and intrinsic goodness. We place high confidence in the technical effort to imbue our machines with that goodness. At the same time, however, and even if our machines were provably and reliably aligned with human morality, delegating responsibility and authority to them would be a monumental decision, affecting every facet of our ability to retain our human relationships, political structures, and individual and collective identities. Nostalgia for the premier status to which our species has recently grown accustomed would be widespread. For some, no matter the eventual trajectory of AI, a world transformed—even for the better—may feel scarcely different from an abrupt ending to our bedrock reality.

A further unsolved question is *Who will decide?* Who will make the choice to delegate, or not delegate, responsibility and authority? Who will give, or withhold, resources? How will any one set of deciders communicate, converge—or come into conflict with—others

attempting to make the same decisions elsewhere? Are we choosing those individuals, those fallible humans, now? Have we, unwittingly, already chosen?

We authors hope to alert these decision-makers, whoever they are, to the decisions that confront us even now and to what may lie ahead. But our aim is not baldly to instill a sense of apprehension about the rise of AI. An abrupt termination in the application of AI's powers could itself precipitate a crisis. Deceleration may be even less politically manageable than our current path, creating significant dangers for those of slower momentum and destroying the hopes of those yearning for further advancements.

Neither blind faith nor unjustified fear can form the basis of an effective strategy; one needs self-doubt to have knowledge but self-confidence to act. Indeed, in the age of AI, this is all the more urgent. We must try to understand the challenges that AI will present even as we lack the prior exposure or the essential experience to guarantee the accuracy of our comprehension. And even as we navigate this daunting task, we must also, to avoid a passive future, surmount the many difficulties already facing our species.

While some may view this moment as humanity's final act, we perceive instead a new beginning. The cycle of creation—technological, biological, sociological, political—is entering a new phase. That phase may operate under new paradigms of, among other things, logic, faith, and time. With sober optimism, may we meet its genesis.